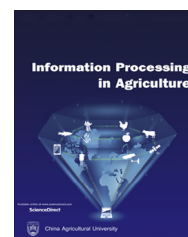




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Tractor path tracking control based on binocular vision



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ABSTRACT

In the process of field operation management, determining how to accurately realize crop row identification and path tracking control is an essential part of tractor automatic navigation. According to the linear operation in the process of cotton field management, the tractor path tracking control system was designed based on binocular vision and the pure pursuit model. A new crop row detection method based on the Census transform and the PID control algorithm with dead zone were used. First, the upper computer software was developed by C++ with the functions of parameter setting and image acquisition and processing. Second, an automatic steering controller was developed based on microprocessor MC9S12XS128 of Freescale. The control program was developed based on modular design using CodeWarrior during development of the PID-based automatic steering control strategy. Finally, a field experiment platform of tractor path tracking control was built, and field experiments under the actual cotton were conducted. The optimal visibility distance was determined by several previous experiments. When the tractor tracks the path with the optimal visibility distance in the growth environment of actual cotton crops, the mean absolute deviation of course angle was 0.95° , and the standard deviation was 1.26° ; the mean absolute deviation of lateral position was 4.00 cm, and the standard deviation was 4.97 cm; the mean absolute deviation of front wheel angle was 2.99° , and the standard deviation was 3.67° . The experimental results show that (1) the crop row detection method based on Census transform can identify the crop line and plan the navigation path well, and (2) the tractor path tracking control system based on binocular vision has good stability and high control precision; thus, the control system can realize the automatic path tracking control of cotton line operation and meets the agricultural requirements of cotton field operation management.

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1. Introduction

The automatic navigation technology of agricultural machinery is an important aspect of precision agriculture. This technology has been widely used in the fields of tillage, sowing, harvesting and other field operations and has greatly

promoted the development of modern agriculture in the direction of high efficiency, low energy and automation [1]. With the continuous improvement of image processing and automatic control technology, the ability to provide accurate real-time information of agricultural machinery makes GPS ideal for automatic control of vehicles. Binocular vision and the pure tracking model can integrate both the absolute and relative positions to provide more reliable and accurate navigation information to guide tractors performing farming operations [2,3].

Recently, in the use of machine vision for crop recognition and processing, stereoscopic matching algorithms are divided into two categories: regional stereo matching algorithm and global stereo matching algorithm.

The global stereo matching algorithm is essentially an optimization algorithm, which transforms the stereo matching problem into an optimization problem for finding the global energy function. The representative algorithms include graph cut algorithm, belief propagation algorithm and collaborative optimization algorithm. The general process is to extract the characteristics of the optical flow field, edge feature, Harris corner feature, SIFT eigenvector, SURF feature and calculate the initial parallax diagram. Then, the global optimization function (dynamic programming, image segmentation, belief propagation, etc.) is used to calculate the disparity value of the image iteratively and optimally. Roviramas et al. [4] used a binocular vision camera to obtain a field parallax image directly, as determined the centerline of crop rows through the linear regression method, and extracted navigation line parameters according to the 3D coordinates. KISE et al. [5] used the cross-correlation matching method to match the global features of the image and established a cosine function model for the quasi cooperating centerline. Torri et al. [6] installed monocular vision navigation systems on a tractor; the lateral deviation was less than 0.024 m, and the heading deviation was less than 1.5° with a tracked speed of 0.25 m/s. Benson et al. [7] designed a monocular vision navigation system with a combined harvester as an experimental platform; the system supports all-weather operation with a precision of GPS level. During the operation at speed of 1.3 m/s, the lateral deviation of the daytime and nighttime navigation was approximately 0.133 m and 0.131 m, respectively. The global algorithm can obtain a lower mismatch rate, but the algorithm is more complex and difficult to meet real-time demand, which is not conducive to practical engineering.

The region stereo matching algorithm is usually done on a set window and matching based on feature vectors extracted from windows. To verify that the use of machine vision technology can improve the steering controller quickly and can correctly obtain the information in crop cultivation operation, Reid et al. [8] installed a color camera in front of the tractor to simulate the forward looking ability of the driver. The line path tracking was realized by identifying the edge features of the crop line as a navigation line. Gerrish et al. [9,10] developed a monocular vision navigation system and conducted a linear crop line tracking test on an electric lawn tractor. Billingsley et al. [11] successfully realized machine vision navigation on a tractor; the lateral deviation was less than 0.02 m when the speed of 1 m/s was followed along a straight line.

Fitzpatrick et al. [12] equipped the windrower with navigation system based on monocular vision successfully, and experimental verification was carried out in alfalfa field. Okamoto et al. [13] installed a monocular vision navigation system for a weeding machine; the root mean square (RMS) of the lateral deviation was less than 3 cm when the line path was tracked in the field. Younse et al. [14] developed a visual navigation system for an automatic spraying robot in a greenhouse; it can realize path tracking and intersection detection. Hague et al. [15] developed an automatic guided vehicle based on the monocular vision navigation system; when the driving speed was 1.5 m/s, the tracking deviation to the line of the curve crop was less than 0.020 m. The region stereo matching algorithm is easy to implement and has high matching accuracy. The most important thing is that the matching accuracy is high and is not affected by the accuracy of feature detection. However, the disadvantage of this method is that the computation of region matching is relatively large and the high false matching rate.

At present, several research studies have been conducted in the field of automatic navigation control based on visual processing. To obtain the navigation path, several researchers have used fuzzy control, PID control and neural network control algorithm to achieve path tracking control [16–24] primarily based on the kinematics model and the dynamic model, and other researchers have developed automatic steering and path tracking control algorithms without a mathematical model [25–28].

As mentioned above, stereo vision techniques can be applied to crop row recognition and serve for path tracking control. Because of the complexity and variability of the field working environment, most former research based on monocular or binocular vision also required reconstruction of the 3D map of the complete field image, which is generally insufficiently accurate and serious time-consuming. In order to reduce the processing time, some research used fewer feature points instead of complete depth images to detect crop rows. Although some corresponding algorithms are capable of detecting crop rows in normal environment, more contrast experiments to check the robustness against different environments are still required. Moreover, after obtaining feature points of crop rows, horizontal strip segmentation and Hough transform are generally used to detect the multiple crop rows. But it is difficult to obtain high-precision stereo matching and cannot fully reflect the advantages of binocular vision. Due to the above disadvantages, it is difficult to obtain a more accurate path tracking control precision, although some path tracking control algorithms such as PID control, fuzzy control and neural network are developed.

In this research, aiming at the low speed linear driving condition of cotton field operation management and taking the tractor as the carrier and hydraulic steering proportional control valve as the control object, a tractor path tracking control system based on binocular vision and pure tracking model was developed. The system used the Census transform to realize crop recognition and path planning based on obtaining the dead reckoning deviation. Based on obtaining the dead deviation, pure tracking model and incremental PID control algorithm, the automatic path tracking control of the tractor was achieved. After completing the hardware

development of the controller and the design of the software system, a field experiment platform was built. Through field experiments in real cotton, the accuracy of crop row recognition and the stability of the path tracking control system and the path planning method were verified.

2. Methods and materials

2.1. Crop row identification and path planning

Setting the centerline of the crop in the middle of the image as the target navigation path, a new crop row detection method based on Census transform was used in this paper [29]. The presented method consisted of three modules that are image preprocessing, feature points detection and crop row detection. Different steps and results of crop row detection based on Census transform are given in Fig. 1, the original color image was obtained by binocular camera (Fig. 1(a)).

Firstly, an improved excess green minus excess red (ExG – ExR) function [30] was used to transform RGB color image to grayscale image (Fig. 1(b)). In order to describe the contour of crop rows, the smallest univalue segment assimilating nucleus (SUSAN) detector was used to detect corner points, which could describe the contour of crop rows.

Then, the module of feature point detection consisted of stereo matching and 3D coordinate point calculation. The accurate stereo matching algorithm based on Census transform was used to calculate the disparity of SUSAN corner point (Fig. 1 (c)). The 3D coordinates of SUSAN corner points were calculated based on the disparities of 2 parallel monocular cameras. If the coordinates in width and height axes of a SUSAN corner point were within the range of width and height thresholds, the point would be extracted to be a feature point of crop rows.

2.1.1. Stereo matching

In order to calculate the 3D coordinates of the feature points, the binocular images were stereo matched to get the disparity values corresponding to the feature points. In this paper, based on the SUSAN corner as the reference point, the parallax value corresponding to the SUSAN corner was calculated by the stereo matching method based on Census transform.

Census transform [31] was an improved algorithm for Rank transform, and it was more robust to image noise. The calculation process was as follows:

- (1) The Census transform mask was constructed, and the center of the mask was placed at the P of the image feature points. The gray value of each pixel in the mask area and the gray value of the center point are compared, the result was compiled into binary Hamming code.
- (2) For corrected binocular images, the SUSAN corners detected in the left image were the reference points. In the right picture, the Hamming code distance between pixel points and reference points was calculated in the same horizontal polar line according to the direction of parallax increase. That was to say, the same number of bits corresponding to the location of Hamming code. The more same number, the shorter distance between Hamming codes.
- (3) Using the SAD measure function to calculate the function for the matching cost and the Hamming code distance of corresponding points was used as matching primitives. The center of the measure mask was traversed by each corresponding point, and minimized the total distance of Hamming code. The obtained corresponding point was the best matching point of the base point p_i . The difference of coordinates in the horizontal direction was the parallax value corresponding to p_i . The specific calculation formula is as follows:

$$\begin{cases} \text{Cost}(i, j, k) = \sum_{(\varepsilon, \eta)} L(i + k + \varepsilon, j + \eta) \\ d = \underset{k}{\operatorname{argmin}} (\text{Cost}(i, j, k)) \end{cases} \quad (1)$$

where $\text{Cost}(i, j, k)$ is matched cost function. (ε, η) is the size of the SAD mask. d is the parallax value corresponding to the p_i .

2.1.2. Three-dimensional coordinate calculation

The camera used in this paper was a binocular camera with parallel optical axes. The 3D coordinates of the feature points can be calculated according to the principle of parallel binocular vision localization.

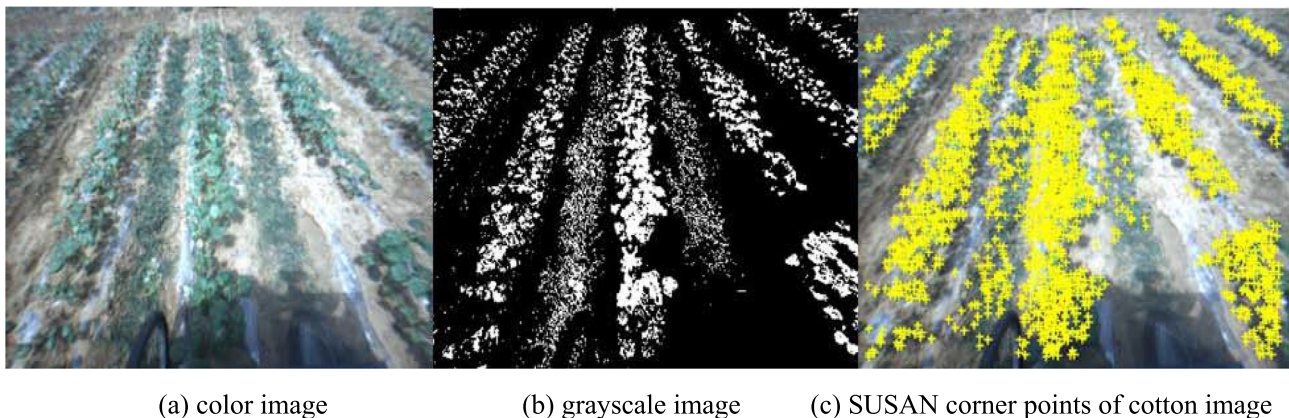


Fig. 1 – Different steps and results of crop row detection based on Census transform.

Taking the left eye camera coordinate system $O_L-X_L Y_L Z_L$ as the reference coordinate system, the spatial coordinates of the feature points were calculated. The directions of X_L , Y_L , and Z_L axes are defined along the width, depth and height of crop rows. Based on the principle of binocular camera, the coordinates of depth, width and height of P_v in the camera coordinate system of left eye can be obtained:

$$Z_L = \frac{bf}{1000\mu_x d} \quad (2)$$

$$\begin{cases} X_L = \frac{Z_L x_i}{f} \\ Y_L = \frac{Z_L y_i}{f} \end{cases} \quad (3)$$

where X_L , Y_L and Z_L are the coordinates of width, height and depth of P_v in the camera coordinate system of left eye, respectively (m). (x_i, y_i) is the coordinates of P_i in the image coordinate system O_i-XY (mm). b is the baseline distance of the binocular camera (mm), and f is the focal length of the camera (mm). d is the disparity value of the projection points in the left and right images (pixel), and μ_x are the physical dimensions of the width direction of the image (mm/pixel).

In order to match the spatial coordinates of the feature points with the actual positions, the left eye camera coordinates should be converted to vehicle coordinate systems $O_v-X_v Y_v Z_v$. As shown in Fig. 2, the coordinate conversion formula is as follows:

$$\begin{bmatrix} X_v \\ Y_v \\ Z_v \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} X_L \\ Y_L \\ Z_L \end{bmatrix} + \begin{bmatrix} 0 \\ h \\ 0 \end{bmatrix} \quad (4)$$

where (X_v, Y_v, Z_v) is the coordinate in the vehicle coordinate system. θ is camera depression angle (deg), and h is camera mounting height (m).

After obtaining the 3D coordinates of SUSAN corner points in vehicle coordinate system, the feature points of crop rows can be extracted according to the elevation difference between crop rows and weeds. In order to reduce the matching calculation, the Census transform mask and SAD measure mask, with a size of 5×5 pixel, were used to match the

SUSAN corners and calculate the 3D coordinates of the SUSAN corner points. Considering the effect of stereo matching error and uneven plant height, to extract enough crop row feature points, experiments set the lower threshold was 0.12 m, and the upper limit of elevation threshold was 0.45 m. The feature points in this threshold interval are feature points of crop line.

Due to the limited vision of the camera, the features of crop line near the left and right edges were blurred. To improve the accuracy of crop line recognition, the effective threshold of crop rows was detected by width threshold. Set the width threshold to ± 1.5 m. The SUSAN corners were stereo matched, their spatial coordinates were calculated, and crop row feature points in elevation and width threshold regions were extracted. Most of the feature points were distributed well-distributed in the crop row area, and the calculated values of the spatial coordinates in the vehicle coordinate system were close to the statistical average. Although a few of weeds were also extracted from crop rows, they were removed from image processing without affecting the accuracy of crop row recognition.

Finally, the amount of crop rows in image were estimated based on the frequency histogram of width distribution. The distance between adjacent crop rows was assigned to the interval, and the range of coordinates in width axis of feature points was assigned to the number of groups in the histogram. The feature points on each crop row were distributed in the shape of a ellipse, and the long axis of the ellipse was similar to the centerline of crop row. After obtaining the amount of crop rows, feature points were used to locate the long axis of their distribution based on the principle component analysis (PCA) method. As shown in Fig. 3, the long axis of the feature points was used to fit the centerline of the corresponding crop row.

2.2. The principle of the tractor path tracking control system

The tractor path tracking control system was mainly composed of an image recognition system, a tracking system and an automatic steering control system. The binocular

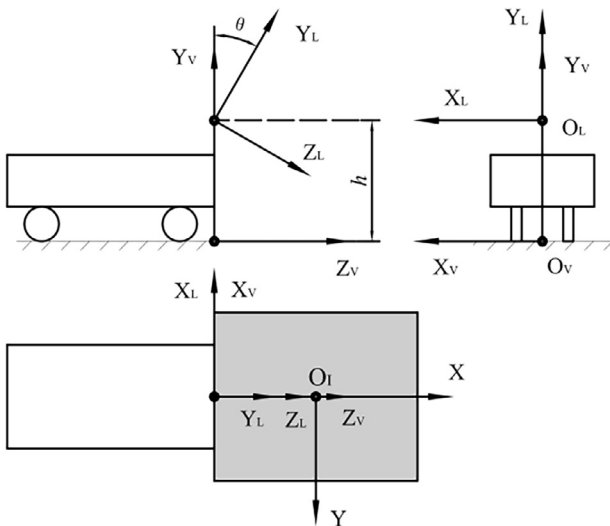


Fig. 2 – Transformation between coordinate systems.

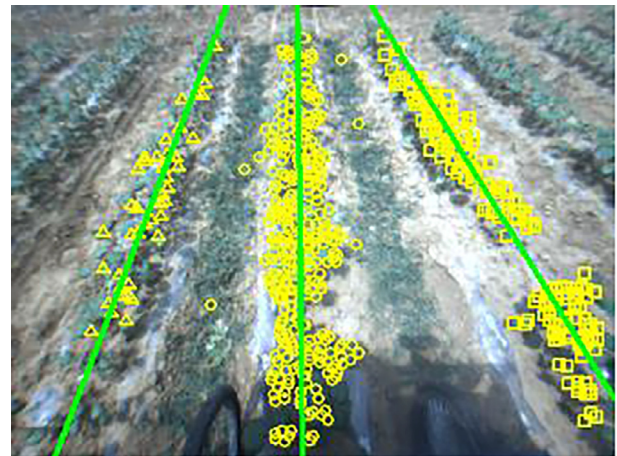


Fig. 3 – Centerlines of crop rows in image.

camera collected the image in real time and sent it to the upper computer by the image acquisition card. The upper monitor processed the dynamic images and identified the crop row centerline and navigation path and later calculated the current tractor driving path dead deviation. The system used a feedback signal to drive a steering proportional control valve and later adjusted the front wheels to achieve path tracking. The operational principle of the control system is shown in Fig. 4.

According to the position of the current driving tractor path and target path, the system calculated the heading angle deviation (φ_e) and lateral position deviation (d_e) between a tractor and a target path, and the dead deviation calculation model was constructed. Fig. 5 shows the geometric relationship between the path of a tractor and target. The current driving path of a tractor coincides with the X_p axis in the path coordinate system $X_p-O_p-Y_p$. When the initial state is $t = 0$, the original point O_p is located in the axle center of the rear axle of the tractor. The following formula can be obtained:

$$y = kx + b \quad (5)$$

where k and b are the slope and the intercept of the target path.

According to the geometric relationship in Fig. 5, the following can be obtained:

$$\begin{cases} \theta_p = \arctan k \\ d_p = b \cos \theta_p \end{cases} \quad (6)$$

The calculation method of (φ_e) and (d_e) were as follows:

$$\begin{cases} \varphi_e = \theta_p - \theta_t \\ d_e = \frac{|kx_{tp} - y_{tp} + b|}{\sqrt{k^2 + 1}} \\ \theta_t = \arctan \left(\frac{y_{tp}}{x_{tp}} \right), t > 0 \end{cases} \quad (7)$$

where θ_p and d_p are the initial heading deviation of the tractor and the initial lateral position deviation, respectively. θ_t is the direction angle of the tractor's driving path at 't' time; when $t = 0$, $\theta_t = 0$. When the tractor is on the right side of the target path, the value of d_e is positive; otherwise, it is negative. x_{tp} and y_{tp} are the front view distances of the tractor corresponding to a coordinate of the path coordinate system $X_p-O_p-Y_p$.

2.3. Calculation of the target angle

Based on the pure tracking model, the target angle of the tractor is calculated. The target angle is determined mainly according to the kinematic equation and the geometric relation of the tractor; the key is to determine the appropriate forward distance. The geometric analysis of the pure tracking model is shown in Fig. 6. The $X-O-Y$ coordinate system is

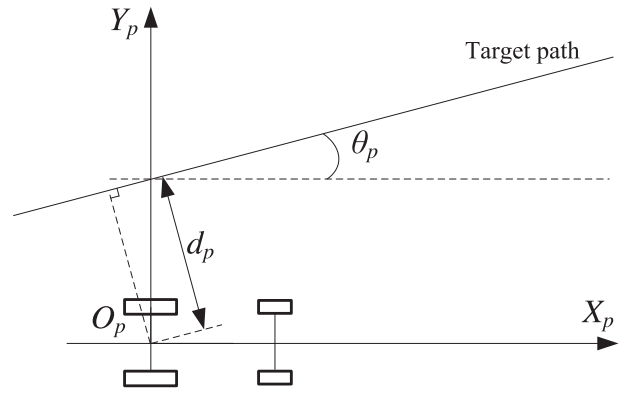


Fig. 5 – Computation model of the deviation of the heading and the position.

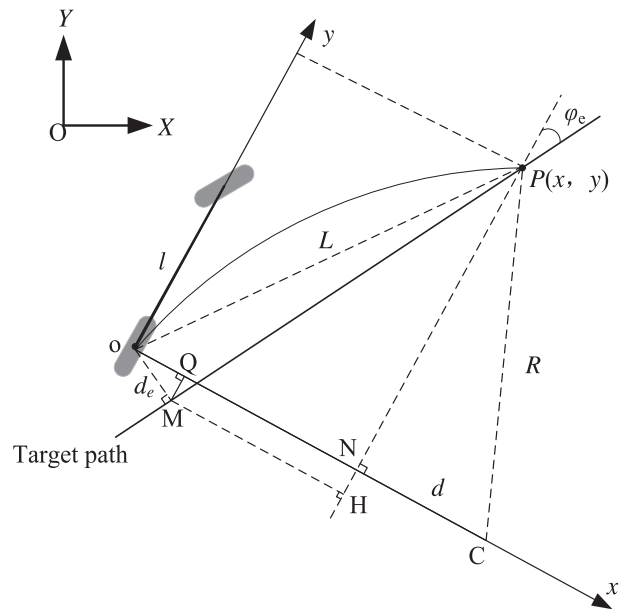


Fig. 6 – Pure tracking model.

the geodetic coordinate system, which takes the axle center of the rear axle of the tractor as the original point and constructs the body coordinate system $x-o-y$. $P(x, y)$ is a point on the target path. The C point is the center of the tractor from the current position point to the target point P trajectory. The radius of the track is R , the front distance of the tractor is L , and the wheelbase is l .

A pure tracking model can be obtained from the geometric relation in Fig. 6:

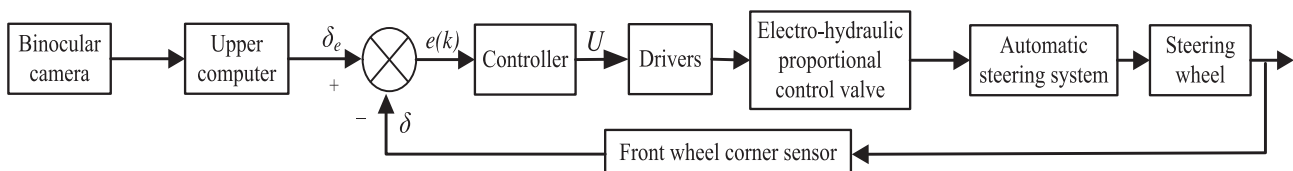


Fig. 4 – Schematic diagram of the path tracking control system.

$$R = \frac{L^2}{2(d_e \cos \varphi_e - \sqrt{L^2 - d_e^2} \sin \varphi_e)} \quad (8)$$

The kinematics of the two wheels of a tractor is as follows:

$$\delta_e = \arctan \frac{l}{R} \quad (9)$$

Thus, the tractor target angle can be represented as

$$\delta_e = \arctan \frac{2l(d_e \cos \varphi_e - \sqrt{L^2 - d_e^2} \sin \varphi_e)}{R} \quad (10)$$

where δ_e is the tractor target angle, φ_e is the heading angle deviation and d_e is the lateral position deviation.

After the values of φ_e , d_e , L , and l are determined, the turning angle of the steering wheel can be calculated. Next, the model compares the turning angle with the actual steering angle of the front wheel, calculates the deviation value of the tractor's rotation angle, brings the deviation value into the control algorithm, and calculates the equivalent control of the steering angle of the front wheel. Finally, the model realizes the control of the steering of the tractor.

2.4. Automatic steering control

The automatic steering control system uses an incremental PID closed loop control with dead-zone. In practical applications, the controller calculates the deviation according to the sampling period, leading to the failure of the continuous integration and differential operation in the PID control algorithm. Taking T as the sampling period, in this paper, the exact integral is approximated by the rectangular integral. Thus, the expression of PID control algorithm after discretization is

$$u(k) = K_p \left[e(k) + \frac{T}{T_I} \int_{j=0}^{k-1} e(j) dt + T_D \frac{e(k) - e(k-1)}{T} \right] + u_0 \quad (11)$$

where $u(k)$ is the controller output, u_0 is the initial value of controller output, K_p is the ratio coefficient, T_D is the differential time constant, and T_I is the integral time constant.

The PID control increment is as follows:

$$\begin{aligned} \Delta u(k) &= u(k) - u(k-1) \\ &= K_p[e(k) - e(k-1)] + K_I e(k) \\ &\quad + K_D[e(k) - 2e(k-1) + e(k-2)] \end{aligned} \quad (12)$$

$$\begin{cases} K_I = \frac{K_p T}{T_I} \\ K_D = \frac{K_p T_D}{T} \end{cases} \quad (13)$$

According to the actual application demand of the automatic steering system, the deviation of the front wheel steering angle is limited. Two variables of maximum deviation ($e_{\max}$) and deviation dead zone ($e_{\min}$) are defined to satisfy the dynamic characteristics of the control system. To increase the response characteristic of the larger deviation, the upper limit ($e_{\max}$) was set. When the deviation was larger than the upper limit, the PID control incremental ($\Delta u(k)$) outputted the maximum value ($\Delta u(k)_{\max}$). When the deviation was less than the upper limit ($e_{\max}$), the control algorithm designed the dead zone ($e_{\min}$) of the control deviation to avoid the oscillation and overheating caused by the system's frequent movement of small deviations, the judgment is shown as

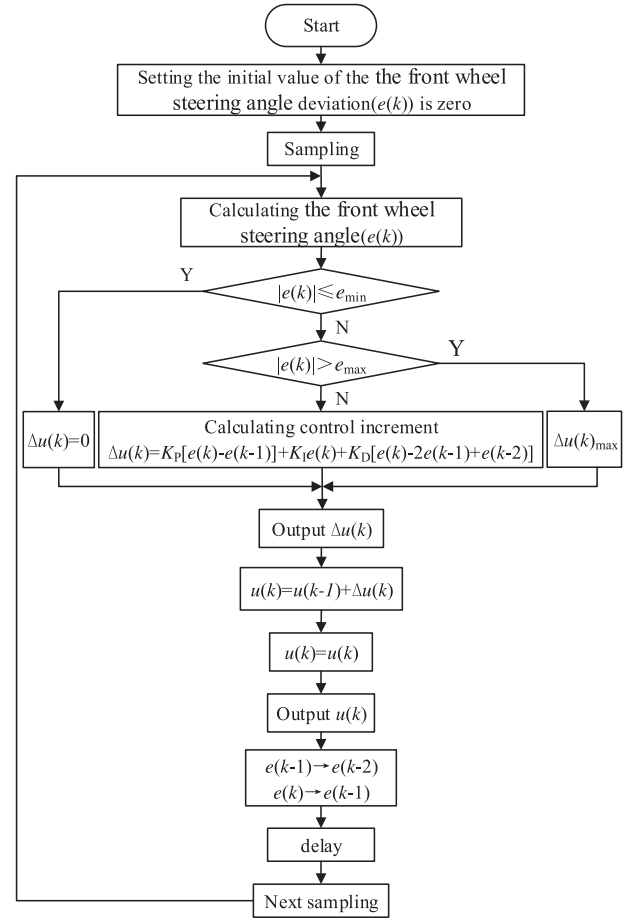


Fig. 7 – PID control program.

$$e(k) = \begin{cases} 0 & |e(k)| \leq e_{\min} \\ e(k) & |e(k)| > e_{\max} \end{cases} \quad (14)$$

The control program flow is shown in Fig. 7.

The control program included the following: PID increment calculation, PID output calculation and control judgment. At the beginning of the control program, the deviation of the steering angle of the front wheel was zero. After the sampling began, the algorithm calculated the actual steering angle of the front wheel and the deviation of the expected angle ($e(k)$). Judging the deviation ($e(k)$) according to the formula (14). When the deviation value was larger than the dead zone, the PID algorithm calculated the control quantity ($u(k)$), thereby limiting the calculated control amount and avoiding the dead zone of the controller. Thus, the output voltage value realized the control of the hydraulic steering proportional control valve.

3. Experiments and results

To verify the performance of tractor tracking control system with binocular vision, experiments were performed using Revò Leopard TG1254 tractor as the supporter equipped with the WYH-3 type hydraulic steering angle sensor ($\pm 45^\circ$, 40 mV/ $^\circ$) and the proportional control valve block (22.8 L/min, 15 MPa). The field test platform for tractor path tracking was built, and the field test was conducted.

3.1. Hardware system

In the tractor path tracking control system based on binocular vision, the hardware system mainly includes upper computer hardware system and controller hardware system. The former part realizes image acquisition, image processing and crop line identification. At the same time, the latter part completes automatic steering control according to the signal sent by the upper computer.

As shown in Fig. 8, the upper computer hardware system for image acquisition was mainly composed of upper computer, 1394 data line, 1394 image acquisition card and binocular camera (Bumblebee 2, produced by Point Grey, Canada). The image resolution acquired by the binocular camera is 640×480 .

Using the PADS software of Mentor Graphics Company, the hardware system of automatic steering controller based on Freescale MC9S12XS128 microprocessor was developed. The controller primarily consisted of the minimum system, the analog signal acquisition, the digital analog conversion output, the serial communication and the CAN bus communication. The controller structure is shown in Fig. 9.

The power supply circuit is composed of the LM2596-ADJ switch voltage regulator, the LM1117 low voltage differential voltage regulator and the filter capacitor to achieve the power supply and voltage regulation of the controller circuit. The analog signal acquisition circuit is based on the LM358 double operational amplifier and is connected to the BAV99 jump line interface circuit to realize the analog signal acquisition. The digital analog conversion output module uses the PM3-5 general I/O port to simulate the SPI interface and is connected to the TLV5618 digital to analog converter to realize the DA output of the control voltage.

3.2. Field experiment of single line bionic cotton

It is important to obtain the optimal visibility distance L that has great influence on the tracking effect through the preliminary test. Therefore, according to the characteristics of the cotton plants, a single line bionic cotton crop line was set up in the field. Next, the optimal visibility distance L with better control response was determined by simulating the low speed linear driving condition in the tractor field operation. Finally, on the basis of determining the optimal visibility



Fig. 8 – Upper computer hardware system.

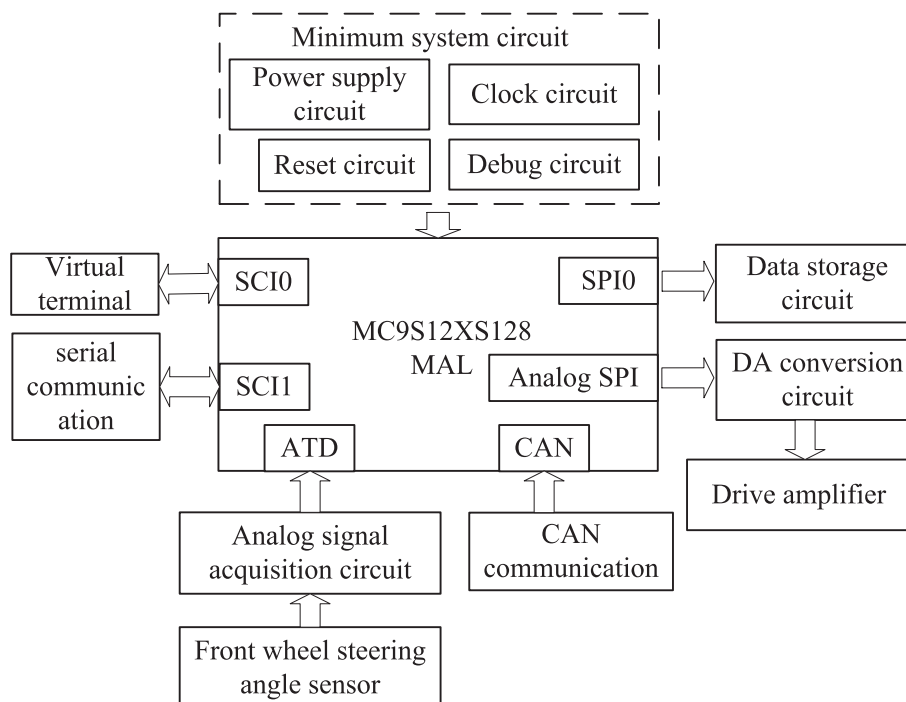


Fig. 9 – Structure of controller.

distance, the field path tracking test of actual cotton was conducted.

Before the experiment, based on the actual environment of the field, the binocular camera was calibrated and the image acquisition processing program was tested. After the binocular camera was installed, the tractor travelled along the crop row. The binocular camera was used to collect the row information of field crops and observed whether the upper computer software system can detect the row crops and plan the navigation path. The experiment showed that the binocular camera works well. The image processing program can identify the line of the crop line and calculate the navigation path. The experimental scene and the image acquisition and processing results are shown in Fig. 10 and Fig. 11, respectively; the lines in Fig. 11 were crop lines identified by the left and right cameras were named the guide lines.

The tractor path tracking test of single line bionic cotton crop was conducted. The experimental conditions were as follows: the speed of the car was 1.20 m/s, the bionic cotton leaves were 0.10 m from the lowest ground height, and the highest height was 0.20 m. After several tests, the path tracking effect of the tractors was better when the visibility distance L was set to 4 m and 5 m. The results of the single line bionic cotton crop tests are shown in Fig. 12.

The average absolute deviation and the standard deviation were used to analyze the course angle deviation, the lateral position deviation and the front wheel steering angle of the tractors in the results. The analysis of the single line bionic cotton crop results are shown in Table 1.

Fig. 12 and table 1 show that when the visibility distance was 5 m, there was a small change in steering angle of front wheel. The mean absolute deviation and standard deviation of the lateral position are far smaller than the test results of the visibility distance of 4 m. The controlled overshoot and the steady state time were small. Although the average absolute deviation of the heading deviation and the standard deviation were relatively large, the control was within 2.50° and fully met the actual operation requirements. In conclusion, under the low speed working condition of the tractor in the field, the optimal visibility distance of the path tracking was 5 m.

3.3. Field experiment of multi-line cotton crops

By several previous tests based on pure tracking model, the optimal visibility distance with good control response was set to $L = 5$ m. The field experiment of multi-line practical cotton crops was conducted according to the actual characteristics of cotton.



Fig. 10 – Field experiment scene.



Fig. 11 – Identification result of the crops' centerline.

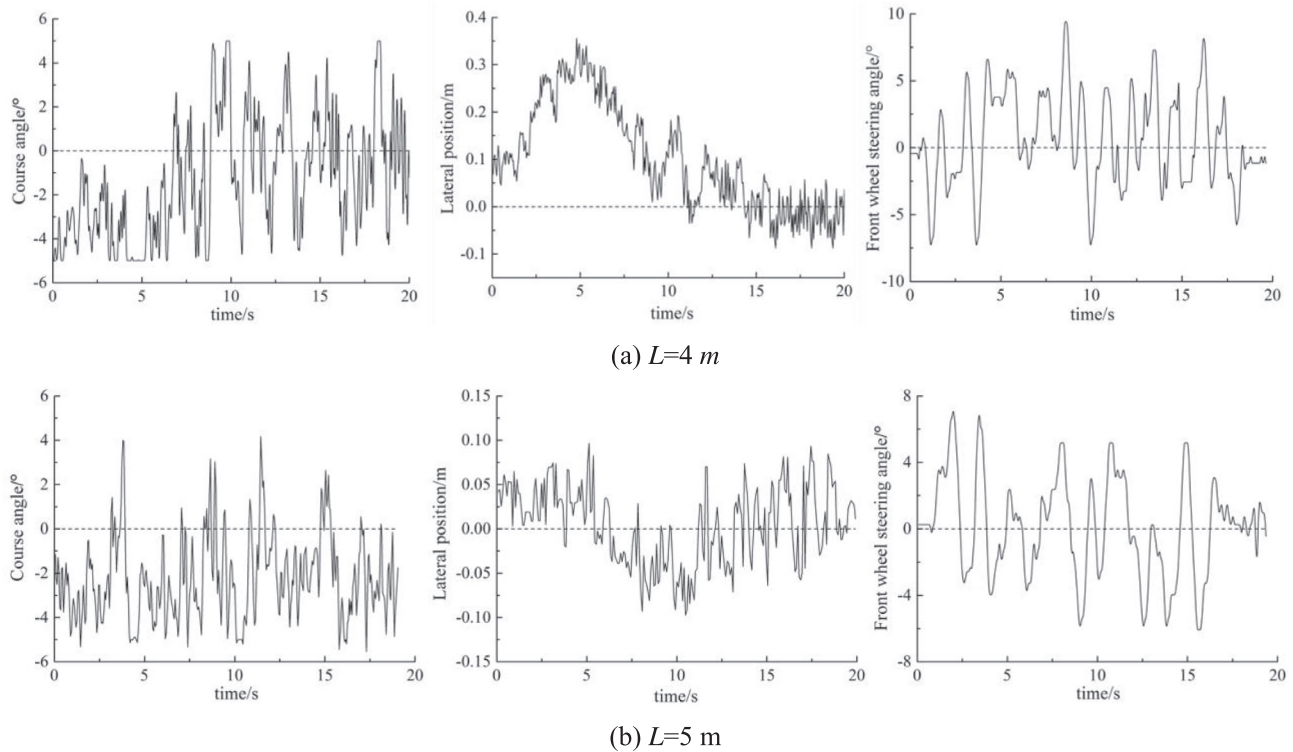


Fig. 12 – Experimental results of bionic cotton crops.

Table 1 – Analysis of experiment results of bionic cotton crops.

Foresight distance (m)	Course angle (°)		Lateral position (cm)		Front wheel steering angle (°)	
	Mean absolute deviation	Standard deviation	Mean absolute deviation	Standard deviation	Mean absolute deviation	Standard deviation
4	1.30	2.94	11	10.80	3.05	3.66
5	2.25	2.17	5	5.04	2.52	3.12

The experimental conditions were as follows: the tractor speed was approximately 1.20 m/s, the cotton crop distance was 0.60 cm, the row spacing was 1.20 m, and the minimum and maximum height of the leaves to the ground was 0.25 m and 0.50 m, respectively. The results of the test are shown in Fig. 13, and the analyses of the test are shown in Table 2.

As shown in Fig. 13 and Table 2, when the foresight distance was 5 m, the mean absolute deviation of the path angle

of the tractors was within 1° , and the standard deviation was within 1.50° . The mean absolute deviation of the lateral position was within 4 cm, and the standard deviation was within 5 cm. The mean absolute deviation of the steering angle of the front wheel was within 3° , and the standard deviation was within 3.70° . The amplitude of fluctuation of the tractors was reduced. The tractor path tracking control system was found to have good stability and high control precision.

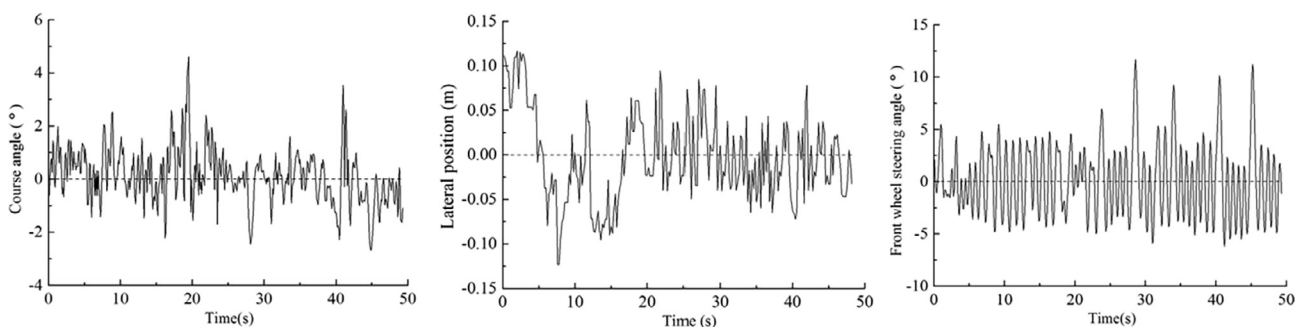


Fig. 13 – Experimental results of actual cotton crops.

Table 2 – Analysis of the experimental results of actual cotton crops.

Foresight distance (m)	Course angle (°)		Lateral position (cm)		Front wheel steering angle (°)	
	Mean absolute deviation	Standard deviation	Mean absolute deviation	Standard deviation	Mean absolute deviation	Standard deviation
5	0.95	1.26	4.00	4.97	2.99	3.67

4. Conclusions

The study was conducted to analyze the demands of linear operation in the process of cotton field management. Through the method of crop row recognition and path planning based on the Census transform, the pure tracking model and the incremental PID control algorithm, a tractor path tracking control system based on binocular vision was developed. The recognition method of the crop line based on the Census transform was found to realize Census transform. The path tracking control system based on binocular vision was observed to have good stability and high control precision, which enables realization of the automatic path tracking control of cotton line operation and meet the agricultural requirements of cotton field operation management.

5. Authors' contributions

Shuo Zhang designed the path tracking control system and the control algorithm, collected the experimental data, wrote the paper, and contributed to the analysis of the experimental data.

Yunyi Wang collected the experimental data, wrote the paper, and contributed to the analysis of the experimental data.

Zhongxiang Zhu designed the algorithm for path tracking control.

Zhen Li collected the experimental data.

Yuefeng Du designed the algorithm for path tracking control.

Enrong Mao wrote the paper and contributed to the analysis of the experimental data.

Conflict of interest

The authors declared that there is no conflict of interest.

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